Pokémon GO

Jay

Katomil

—— Raid PvP

- **Curveball** Curveball **1.7** —— Razz Berry 1.5
- —— Excellent
- **Nice / Great / Excellent** = Great——Excellent

+ Berry +

Curveball	1.7×
Nice throw	1.0–1.3×
Great throw	1.3–1.7×
Excellent throw	1.7–2.0×
Razz Berry	1.5×
Silver Pinap Berry	1.8×
Golden Razz Berry	2.5×
Great Ball	1.5×
Ultra Ball	2.0×

Curveball Excellent + Golden Razz + Ultra Ball = $1.7 \times 2.0 \times 2.5 \times 2.0 =$ **17**

Berry

Berry			
Razz Berry	1.5×		" "Berry
Golden Razz Berry	2.5×		Shiny——
Pinap Berry	×2		
Silver Pinap Berry	1.8×	+ 2.33×	——
Nanab Berry			

Raid **Team GO Rocket** Shiny 100% —— Pinap Razz Shiny Golden Razz

3 Community Day

1.
2. Berry/
3. Curveball
4.
5. ——

	100
	125
Haunter	300
Gengar	500
Star Piece	× 1.5

	100
Curveball	+20
Nice throw	+20
Great throw	+100
Excellent throw	+1,000
	+50
	+1,000
Lucky Egg	× 2

1.
2. **5** ——
3. **25%** 125 100

”Windy “ / /

Adventure Sync app app Google Fit / Apple Health

- —— app
- **Buddy** —— app
- 9 25km 5km + 50km 10km + + Rare Candy 50km

Incense ~5 ~1 200 15 —— ~30

Lure PokéStop 30 Lure

——

CP

1–29		+ 5
30+	30	35

30 IV 30 15 25 PokéStop 25

+5 —— 20 25

+ 10 50 Best Buddy 51 40 XL Candy 31

20	Raid	20	25
30	——		
31	XL Candy 40→50		
35	Wayfarer	PokéStop	
40	XL Candy		

30 30 XL Candy

CP

CP **CP** ——

CP

$$\mathrm{CP} = \left\lfloor \frac{(\mathrm{Atk}_{\mathrm{base}} + \mathrm{Atk}_{\mathrm{IV}}) \times \sqrt{\mathrm{Def}_{\mathrm{base}} + \mathrm{Def}_{\mathrm{IV}}} \times \sqrt{\mathrm{Sta}_{\mathrm{base}} + \mathrm{Sta}_{\mathrm{IV}}}}{10} \times \mathrm{CPM}^2 \right\rfloor$$

1. —— Machamp 234 Blissey 129

2. **IV** —— 0–15
3. **CPM CP** —— 1–51 = CPM = CP

+1 IV CP +1 +1 PvP

IV

0 15

- **IV** 0–15
- **IV** 0–15
- **HP** / **IV** 0–15

15/15/15 "Hundo" 100% →

IV

IV 150–250+ 0–15

Machamp —— 234 / 159 / 207

- 0 IV → = 234 15 IV → = 249
- 15/234 = **6.4%** IV

Blissey —— 129 / 169 / 496

- 0 IV → = 129 15 IV → = 144
- 15/129 = **11.6%** IV
- 15/496 = **3.0%** HP IV

IV 230+ IV 120 IV

```
import matplotlib.pyplot as plt
import numpy as np

pokemon = {
    'Machamp': {'atk': 234, 'def': 159, 'sta': 207},
    'Blissey': {'atk': 129, 'def': 169, 'sta': 496},
}

fig, ax = plt.subplots(figsize=(10, 5))

x = np.arange(3)
width = 0.35
```

```python
stats = ['Attack', 'Defense', 'Stamina']

machamp_pct = [15/234*100, 15/159*100, 15/207*100]
blissey_pct = [15/129*100, 15/169*100, 15/496*100]

bars1 = ax.bar(x - width/2, machamp_pct, width, label='Machamp (234/159/207)',
               color='#e74c3c', edgecolor='white', linewidth=0.5)
bars2 = ax.bar(x + width/2, blissey_pct, width, label='Blissey (129/169/496)',
               color='#ff69b4', edgecolor='white', linewidth=0.5)

for bar, val in zip(bars1, machamp_pct):
    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.2,
            f'{val:.1f}%', ha='center', va='bottom', fontsize=11, fontweight='bold')
for bar, val in zip(bars2, blissey_pct):
    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.2,
            f'{val:.1f}%', ha='center', va='bottom', fontsize=11, fontweight='bold')

ax.set_ylabel('% Stat Increase from 15 IV', fontsize=12)
ax.set_title('IV Impact: 15 IV as % of Total Stat', fontsize=14, fontweight='bold')
ax.set_xticks(x)
ax.set_xticklabels(stats, fontsize=12)
ax.legend(fontsize=11)
ax.set_ylim(0, max(max(machamp_pct), max(blissey_pct)) + 2)
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)
plt.tight_layout()
plt.show()
```

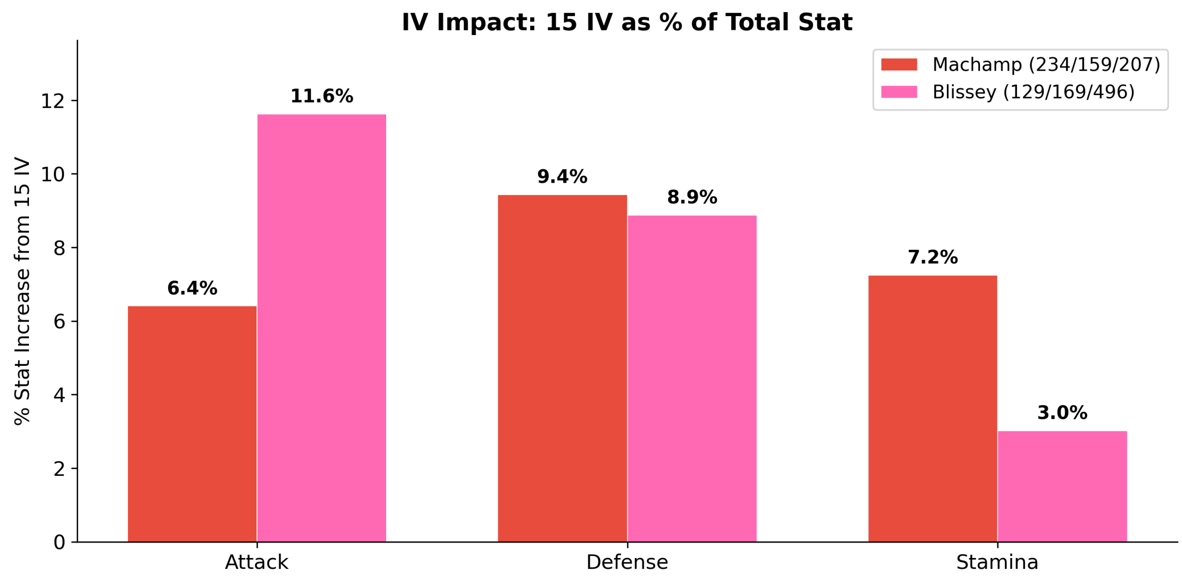

Figure 1: Same 15 IV points, very different impact. IVs are a bigger deal where the base stat is low.

" "____ " "Pokémon GO

$$\text{Damage} = \left\lfloor \frac{1}{2} \times \text{Power} \times \frac{\text{Atk}_{\text{effective}}}{\text{Def}_{\text{effective}}} \times \text{STAB} \times \text{Effectiveness} \right\rfloor + 1$$

____

```
import matplotlib.pyplot as plt
import math

cpm_40 = 0.7903

# Machamp using Counter (12 power, Fighting) vs Tyranitar (Dark/Rock)
# Tyranitar: base def 207
machamp_atk_15 = (234 + 15) * cpm_40   # 196.8
machamp_atk_0  = (234 + 0)  * cpm_40   # 184.9
ttar_def = (207 + 15) * cpm_40          # 175.4

stab = 1.2         # Counter is Fighting, Machamp is Fighting
se = 1.6            # Fighting vs Dark/Rock
```

```
neutral = 1.0
power_counter = 12
power_rock_slide = 50  # neutral move for comparison

def damage(power, atk, defn, stab_mult, eff):
    return math.floor(0.5 * power * (atk / defn) * stab_mult * eff) + 1

# Scenario 1: Perfect IVs vs zero IVs (same move, same matchup)
dmg_15iv = damage(power_counter, machamp_atk_15, ttar_def, stab, se)
dmg_0iv  = damage(power_counter, machamp_atk_0,  ttar_def, stab, se)

# Scenario 2: Super effective vs neutral (same pokemon, same IVs)
dmg_se      = damage(power_counter, machamp_atk_0, ttar_def, stab, se)
# Use Rock Slide (Rock, 50 power) - not STAB, not SE vs Tyranitar (resisted actually)
# Let's instead compare: Counter (Fighting, SE) vs a hypothetical neutral fast move
dmg_neutral = damage(power_counter, machamp_atk_15, ttar_def, stab, neutral)

# Scenario 3: Shadow vs normal (15 IV both)
shadow_atk = machamp_atk_15 * 1.2
dmg_shadow = damage(power_counter, shadow_atk, ttar_def, stab, se)
dmg_normal = damage(power_counter, machamp_atk_15, ttar_def, stab, se)

fig, axes = plt.subplots(1, 3, figsize=(14, 5))

# Plot 1: IV comparison
colors1 = ['#ef4444', '#fbbf24']
bars1 = axes[0].bar(['15/15/15\nIVs', '0/0/0\nIVs'], [dmg_15iv, dmg_0iv], color=colors1,
                     edgecolor='white', linewidth=0.5, width=0.5)
for bar, val in zip(bars1, [dmg_15iv, dmg_0iv]):
    axes[0].text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.2,
                 str(val), ha='center', va='bottom', fontsize=16, fontweight='bold')
axes[0].set_title('IVs: 15 vs 0 Attack\n(Counter vs Tyranitar)', fontsize=11, fontweight='bo
axes[0].set_ylabel('Damage per Hit', fontsize=11)
pct_diff_iv = abs(dmg_15iv - dmg_0iv) / dmg_0iv * 100
axes[0].text(0.5, 0.5, f'Δ = {pct_diff_iv:.0f}%', transform=axes[0].transAxes,
             ha='center', fontsize=14, color='gray', fontstyle='italic')

# Plot 2: Type effectiveness
colors2 = ['#22c55e', '#94a3b8']
bars2 = axes[1].bar(['Super\nEffective', 'Neutral'], [dmg_se, dmg_neutral], color=colors2,
                     edgecolor='white', linewidth=0.5, width=0.5)
for bar, val in zip(bars2, [dmg_se, dmg_neutral]):
```

```
    axes[1].text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.2,
                 str(val), ha='center', va='bottom', fontsize=16, fontweight='bold')
axes[1].set_title('Type: Super Effective vs Neutral\n(0 IV SE vs 15 IV Neutral)', fontsize=1
pct_diff_type = (dmg_se - dmg_neutral) / dmg_neutral * 100
axes[1].text(0.5, 0.5, f'Δ = {pct_diff_type:.0f}%', transform=axes[1].transAxes,
             ha='center', fontsize=14, color='gray', fontstyle='italic')

# Plot 3: Shadow vs normal
colors3 = ['#8b5cf6', '#3b82f6']
bars3 = axes[2].bar(['Shadow', 'Normal'], [dmg_shadow, dmg_normal], color=colors3,
                     edgecolor='white', linewidth=0.5, width=0.5)
for bar, val in zip(bars3, [dmg_shadow, dmg_normal]):
    axes[2].text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.2,
                 str(val), ha='center', va='bottom', fontsize=16, fontweight='bold')
axes[2].set_title('Shadow Bonus\n(15/15/15 both)', fontsize=11, fontweight='bold')
pct_diff_shadow = (dmg_shadow - dmg_normal) / dmg_normal * 100
axes[2].text(0.5, 0.5, f'Δ = {pct_diff_shadow:.0f}%', transform=axes[2].transAxes,
             ha='center', fontsize=14, color='gray', fontstyle='italic')

for ax in axes:
    ax.spines['top'].set_visible(False)
    ax.spines['right'].set_visible(False)
    ax.set_ylim(0, max(dmg_shadow, dmg_se) * 1.2)

fig.suptitle('What Actually Changes Your Damage - Machamp Counter vs Tyranitar (Level 40)',
             fontsize=13, fontweight='bold', y=1.02)
plt.tight_layout()
plt.show()
```

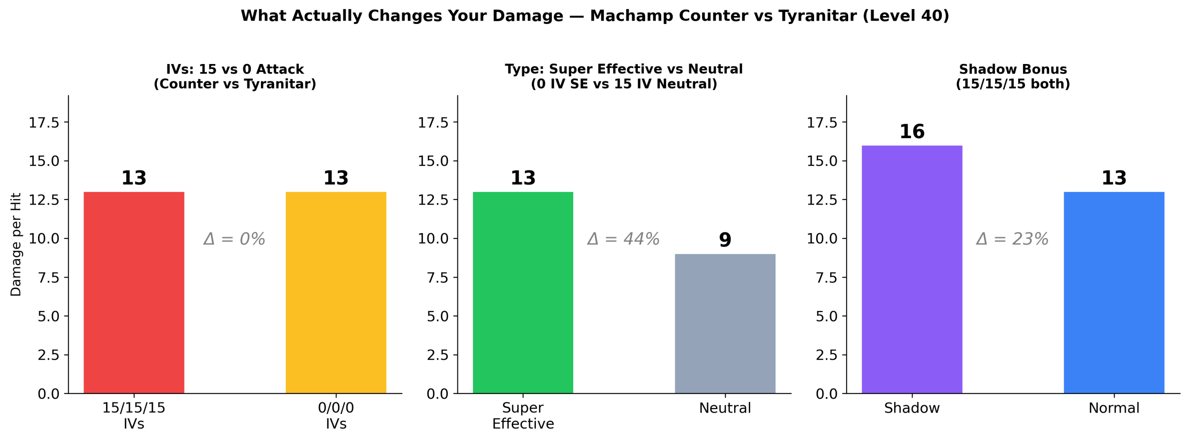

Figure 2: Three scenarios showing what actually changes damage output: IVs barely move the needle, type effectiveness dominates everything.

0 IV Machamp 15 IV Machamp ——IV 5%

IV

→ / / HP = Hundo

PokeGenie **CalcyIV** app IV PvP 4096 IV

IV—— 500 1

IV Shadow

18 ——

	1.6×
	0.625×
	2.56×
	0.391×

Pokémon GO —— 0.391×

STAB **1.2** STAB —— = 1.2 × 1.6 = **1.92×**

Shadow Purified

Team GO Rocket Giovanni **Shadow**

Shadow

Shadow

- Shadow = × **1.2** 20%
- Shadow = × **0.833** 20%

20%

Shadow IV

Machamp 40 CPM = 0.7903

15 IV	$(234 + 15) \times 0.79 = 196.8$	
0 IV	$(234 + 0) \times 0.79 = 184.9$	-6.0%
Shadow 0 IV	$(234 + 0) \times 0.79 \times 1.2 =$ **221.9**	**+12.8%**
Shadow 15 IV	$(234 + 15) \times 0.79 \times 1.2 =$ **236.1**	**+20.0%**

IV Shadow Machamp IV Machamp Shadow IV 0 15 3 IV Shadow

```
import matplotlib.pyplot as plt
import numpy as np

base_atk, base_def, base_sta = 234, 159, 207
iv = 13
pure_iv = 15
cpm = 0.7903

forms = ['Shadow\n(13/13/13)', 'Normal\n(13/13/13)', 'Purified\n(15/15/15)']
colors = ['#8b5cf6', '#3b82f6', '#22c55e']

vals = {
    'Attack': [
        (base_atk + iv) * cpm * 1.2,
        (base_atk + iv) * cpm,
        (base_atk + pure_iv) * cpm,
    ],
    'Defense': [
        (base_def + iv) * cpm * 0.833,
        (base_def + iv) * cpm,
        (base_def + pure_iv) * cpm,
    ],
    'Stamina': [
        int((base_sta + iv) * cpm),
        int((base_sta + iv) * cpm),
```

```
        int((base_sta + pure_iv) * cpm),
    ],
}

fig, axes = plt.subplots(1, 3, figsize=(13, 5))

for ax, (stat_name, stat_vals) in zip(axes, vals.items()):
    bars = ax.bar(forms, stat_vals, color=colors, edgecolor='white', linewidth=0.5)
    for bar, val in zip(bars, stat_vals):
        label = f'{val:.1f}' if isinstance(val, float) else str(val)
        ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 1,
                label, ha='center', va='bottom', fontsize=11, fontweight='bold')
    ax.set_title(stat_name, fontsize=13, fontweight='bold')
    ax.spines['top'].set_visible(False)
    ax.spines['right'].set_visible(False)
    ymin = min(stat_vals) * 0.88
    ymax = max(stat_vals) * 1.08
    ax.set_ylim(ymin, ymax)

fig.suptitle('Shadow vs Normal vs Purified - Machamp at Level 40',
             fontsize=14, fontweight='bold', y=1.02)
plt.tight_layout()
plt.show()
```

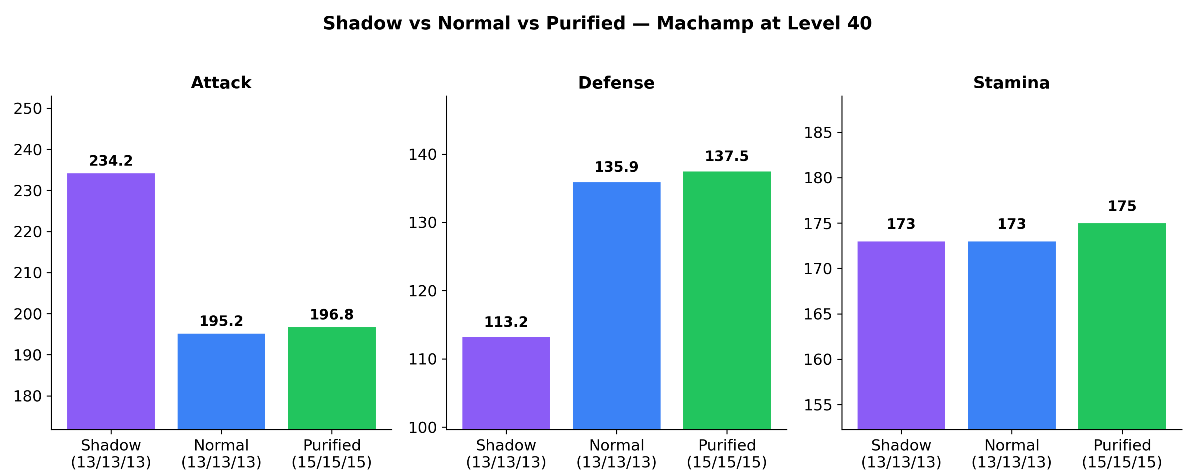

Figure 3: Shadow vs Normal vs Purified — Machamp with 13/13/13 IVs (becomes 15/15/15 after purifying). Level 40.

Purified 15/15/15 Normal 13/13/13 IV +2 Shadow

- Shadow 1.0× 1.0×
- **IV +2** 15
- 20%
- 25
- **Return** PvP

20%

IV 13/13/13	+2 → 15/15/15 Hundo Shadow Golbat Raticate Hundo → ——1 Zubat Aron Starly Shadow
Mega Primal	Shadow Mega Primal Mega Gengar Mega Charizard Primal Groudon—— Shadow
Return	PvP Return Purified Sableye Great League

Shadow Raid 13+ IV Shadow Hundo

Raid

Raid Boss

Raid

1		Rare Candy
3		
5	—— 3–6+	—— Raid
Mega	—— 3–6+	Mega Mega

Elite / Primal	—— 5–10+	Primal Groudon/Kyogre

Raid

- 25 40 Boss Counter
- **Pokégenie** —— app Raid
- **Remote Raid Pass**
- **Raid 20** 25 IV 10+ 10/10/10
- **Rare Candy** Rare Candy

&

Forever Forward

2026 6 2 9 8

	Scenic Sunday	Incense Buddy 3 Mateo	—— Buddy
	Max Monday	Power Spots Power Spots 6 AM – 9 PM	/Max
	Showcase Tuesday	5 PokéStop	——
	Raid Hour	5 Boss 6–7 PM	—— Raid
	GO Battle Thursday	PvP 4 10 50 5	——PvP
	Friendship Friday	2 Lucky 10% 2 XL Candy 31 +	——
	Community Day		

9:00 AM	Adventure Sync	25/50 km
6–7 PM	Spotlight Hour	+

Community Day

- ——
- —— 5
- / —— 3 Star Piece 4.5

Community Day

Buddy

Buddy 1km 3km 5km 20km

Buddy

Buddy

Good Buddy	1	Buddy
Great Buddy	70	——Buddy
Ultra Buddy	150	
Best Buddy	300	**+1** Buddy

Best Buddy ~30 +1 CP ——

Buddy Buddy

&

Raid

Good Friend	1	Raid +1
Great Friend	7	Raid +2
Ultra Friend	30	Raid +4

Best Friend	90		Raid +5 **Lucky Friend**

Best Friend **Lucky Egg** 200,000 100k $\times$ 2 Lucky Egg

- Lucky
- **IV** IV = IV
- Shiny " "

Lucky

Lucky IV 12/12/12 —— Lucky

1. **Lucky Friend** —— Best Friend Lucky Friend Lucky
2. —— **2020** Lucky **45** **Lucky** ——
3. —— Lucky

50 Lucky

PokéStop Adventure Sync

2 km	PokéStop		
5 km	PokéStop		
7 km			
10 km	PokéStop		
12 km	Rocket		
5/10 km	Adventure Sync	9 AM	——

- 3 2km 10km 12km
- IV
- IV **10/10/10** —— Raid IV
- Adventure Sync app

&

- 100–500 +25%
- 10km
- GBL Go Battle League
- Adventure Sync 50km =
- Star Piece 30 1.5 —— Community Day

- Raid PvP
- 10,000–100,000
- Lucky —— Lucky
- XL Candy 50 40→50 ~300,000

- 3 1 Buddy Rare Candy
- **XL Candy** 31 40 100 → 1 XL

Rare Candy Raid GBL ——

Ultra Ball	
Pinap Berry	
Golden Razz Berry	Nanab Berry
Max Potion Max Revive	
Star Piece Lucky Egg	Lure
Raid Pass	

Hundo	15/15/15 IV 100% /
Nundo	0/0/0 IV
Shiny	Raid Rocket Shiny —— Shiny
Shundo	Shiny + Hundo Shiny 15/15/15 ——
Shadow	Rocket 1.2× / 0.833×
Purified	Shadow IV +2 Mega
Lucky	IV 12/12/12
STAB	—— 1.2
CMP	Charged Move Priority ——
Farming down	Fast Move
Spice pick	
XL Candy	31 + 40
CPM	CP Multiplier——CP